

SECURECORP

Cybersecurity Assessment Capstone

Junior Cybersecurity Consultant — Final Submission Report

Assessment scope: Controlled training laboratory only

Environment: WSL Ubuntu attacker/analyst • Metasploitable2 + DVWA target • Wazuh Cloud monitoring

Prepared as a professional capstone submission using the evidence actually produced during the lab.

1. Executive Summary

This engagement assessed a controlled SecureCorp training environment from both offensive and defensive perspectives. The completed work established the lab, identified the target attack surface with Nmap, validated DVWA availability, and deployed a Wazuh Cloud agent on the Ubuntu WSL analyst environment. The target exposed a broad set of network services, including FTP, SSH, HTTP, MySQL, PostgreSQL, VNC and X11, creating a significant attack surface.

The evidence package deliberately distinguishes verified activities from activities that were not successfully executed. No successful compromise is claimed without supporting evidence. This preserves the integrity of the assessment and prevents unsupported conclusions.

Overall conclusion: the lab demonstrates a high-risk legacy target profile and a functioning centralized monitoring environment, while target-side Wazuh telemetry and several application/credential attack exercises remain unverified in the captured evidence.

Recommended priorities are network segmentation/firewalling, removal of unnecessary exposed services, patching/modernization, stronger authentication and MFA where applicable, application security controls, and complete target-side logging/monitoring.

2. Scope and Rules of Engagement

Item	Definition
Authorized scope	Only the SecureCorp lab assets controlled for the capstone: WSL Ubuntu, Metasploitable2/DVWA and Wazuh Cloud.
Testing rule	No public, third-party or shared systems were intentionally targeted.
Credential testing	Password attacks are restricted to self-created low-privilege lab accounts.
Evidence rule	Findings are reported only when supported by terminal output, dashboard evidence, screenshots or saved scan results.
Assessment loop	Build → Discover → Attack → Log → Detect → Investigate → Assess → Remediate → Report

3. Lab Architecture

The final working architecture consisted of an Ubuntu WSL environment acting as attacker/analyst, a Metasploitable2 machine hosting DVWA as the vulnerable target, and Wazuh Cloud as the monitoring platform. The Wazuh agent that was successfully enrolled was installed on the WSL Ubuntu environment. Target-side Wazuh agent/log ingestion was not established in the captured evidence and is therefore not represented as completed.

Component	Role	Observed evidence
WSL Ubuntu 26.04	Attacker / analyst	Wazuh agent installed and connected; local interface observed.
Metasploitable2	Vulnerable target	DVWA reachable; Nmap identified 17 open services in the captured scan.
DVWA	Web application target	Login/application access verified at the target HTTP service.
Wazuh Cloud	Central monitoring	Environment Ready; agent shown Active in dashboard.

Evidence: Wazuh environment and agent screenshots are included in the appendix.

4. Asset Inventory

Asset	Type / Role	Security relevance
Metasploitable2	Linux vulnerable VM / target	Primary attack surface; exposes legacy services.
DVWA	Web application	Application-security testing target for brute force, SQLi, XSS and command injection.
WSL Ubuntu	Linux attacker/analyst	Runs reconnaissance/testing tools and hosts Wazuh agent.
Wazuh Cloud	Monitoring/SIEM platform	Centralizes agent health and security telemetry.
Lab network	Connectivity	Enables communication between attacker and target.
Evidence files	Security records	Screenshots, scan outputs and report artifacts support auditability.

5. CIA Analysis

Asset	Confidentiality	Integrity	Availability
Metasploitable2	High	High	High
DVWA	High	High	Medium
Wazuh Cloud	High	High	High
Lab credentials	High	High	Medium
Security evidence	High	High	Medium

6. Network Discovery — Nmap

The recorded service-version scan identified the following target services. This is the strongest completed technical discovery artifact in the evidence set.

Port	Service / Version	Assessment note
21/tcp	vsftpd 2.3.4	Legacy FTP; high exposure if unnecessary.
22/tcp	OpenSSH 4.7p1	Legacy SSH service; authentication surface.
23/tcp	Linux telnetd	Clear-text remote administration; high risk.
25/tcp	Postfix smtpd	Mail service exposed.
53/tcp	ISC BIND 9.4.2	Legacy DNS service.
80/tcp	Apache 2.2.8	Web/DVWA exposure.
139/445	Samba	File-sharing attack surface.
2049/tcp	NFS	Network file-system exposure.
2121/tcp	ProFTPD 1.3.1	Second FTP service exposed.
3306/tcp	MySQL 5.0.51a	Database exposed to attacker network; high priority.
5432/tcp	PostgreSQL 8.3	Database service exposed.
5900/tcp	VNC	Remote desktop exposure.
6000/tcp	X11	Legacy graphical service exposure.

The captured scan also reported rpcbind, login and shell-related services. The overall result is a broad legacy attack surface with substantially more services exposed than a hardened production host should normally require.

7. Application Security Assessment

DVWA access was successfully established, providing a valid application-testing target. The handbook requires testing of brute force, SQL injection, XSS and command injection. The evidence captured so far confirms application access, but does not contain sufficient proof of successful exploitation for each individual module. Those exercises are therefore marked as not verified rather than fabricated.

Activity	Status	Evidence / conclusion
DVWA application mapping	Completed	DVWA login/access verified.
DVWA Brute Force	Not verified	No reliable successful-attempt evidence captured.
SQL Injection	Not verified	No captured output proving additional data disclosure.
Cross-Site Scripting	Not verified	No captured browser evidence proving script execution.
Command Injection	Not verified	No captured harmless OS-command result.
FTP brute force	Not verified	FTP listener was observed, but the client session was closed by the service before reliable login testing.
SSH brute force	Not verified	Application-level SSH connection was refused during the recorded test.

8. Wazuh Monitoring and Detection

Wazuh Cloud was successfully created for the capstone and the Ubuntu WSL agent was enrolled. The dashboard evidence shows one active agent named SecureCorp-WSL. The agent state was also checked locally and reported connected.

Important scope limitation: the successful Wazuh agent was on WSL Ubuntu, not on Metasploitable2. Consequently, this report does not claim automatic detection of target-side SSH/FTP/DVWA attacks unless target-side evidence exists.

Monitoring item	Status	Assessment
Wazuh Cloud environment	Verified	Environment created and accessible.
Wazuh agent enrollment	Verified	SecureCorp-WSL shown Active.
Agent connectivity	Verified	Local state reported connected.
Target-side agent	Not verified	No evidence of Wazuh agent on Metasploitable2.
Automatic detection of Week 2 attacks	Not visible	Cannot be claimed without target-side telemetry and corresponding alerts.

9. Risk Assessment

Risk	Threat / Vulnerability	Impact	Priority
Critical	Command injection in a vulnerable web application	Potential OS-level compromise	Immediate
High	Legacy exposed services and weak authentication surfaces	Unauthorized access / lateral movement	High
High	Exposed MySQL/PostgreSQL services	Database disclosure or abuse	High
High	Telnet/VNC/X11 exposure	Remote access and credential/session compromise	High
High	Insufficient target-side monitoring	Delayed detection and investigation	High
Medium	Application-level XSS/SQLi if present	Data exposure / session compromise	High

10. Recommendations and Remediation Plan

Area	Recommendation	Priority
Network exposure	Restrict inbound access with host/network firewalls; expose only required services.	Critical
Legacy services	Remove or isolate Telnet, VNC, X11, unused FTP and unnecessary RPC/NFS services.	Critical
Patching	Replace/upgrade legacy Apache, OpenSSH, database, FTP and other obsolete components.	Critical
Authentication	Use strong passwords, lockout/rate limiting and MFA where supported.	High
Web security	Validate/sanitize input, use parameterized queries, output encoding and safe command execution.	High
Monitoring	Install Wazuh agent/log collection on the target and ingest Linux auth + Apache/DVWA logs.	High
Detection	Create/validate rules for SSH/FTP brute force and web attack indicators; retain timestamps and source IPs.	High
Backups	Maintain tested backups and recovery procedures for critical data/configuration.	Medium
Awareness	Train users/admins on credential hygiene, phishing resistance and incident reporting.	Medium

11. Incident / Investigation Conclusion

Based on the evidence available, no confirmed compromise of the Metasploitable2 target is asserted. The engagement did establish a vulnerable legacy target, a working DVWA application, a broad exposed service set, and a functioning Wazuh Cloud agent on the analyst environment.

For a production-quality continuation, the next investigation cycle should first establish target-side Wazuh telemetry, then replay each Week 2 activity with timestamps and source/target IPs. Each action should be classified as Event, Alert or Incident according to the handbook, with an evidence screenshot and corresponding Wazuh rule/log reference.

Detection status summary: Automatic = only where an actual alert is evidenced; Manual = where analyst review of logs is required; Not Visible = where the relevant target-side telemetry was not captured. No detection claim is made merely because Wazuh Cloud exists.

12. Evidence Index

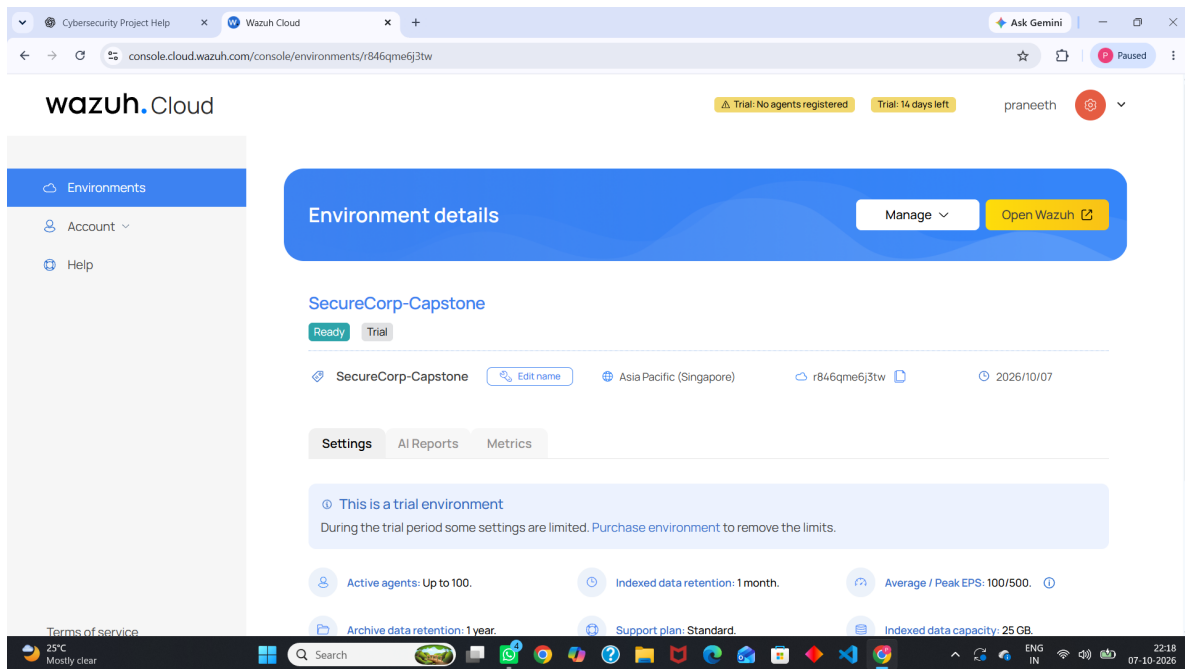
Evidence	Purpose	Status
W1-T1.1-DVWA.png	DVWA target access / application availability	Captured
W2-T2.1-nmap-services.txt	Nmap service/version discovery	Captured
W1-T1.5-wazuh-agent-active.png	Wazuh Cloud agent Active	Captured
Wazuh environment screenshot	SecureCorp-Capstone environment	Captured
Wazuh deployment screenshot	Linux agent deployment configuration	Captured
Wazuh overview screenshot	Monitoring dashboard state	Captured
DVWA brute force evidence	Required attack evidence	Not verified
SQLi/XSS/Command Injection evidence	Required application attack evidence	Not verified
FTP/SSH brute force evidence	Required credential attack evidence	Not verified

13. Final Submission Checklist

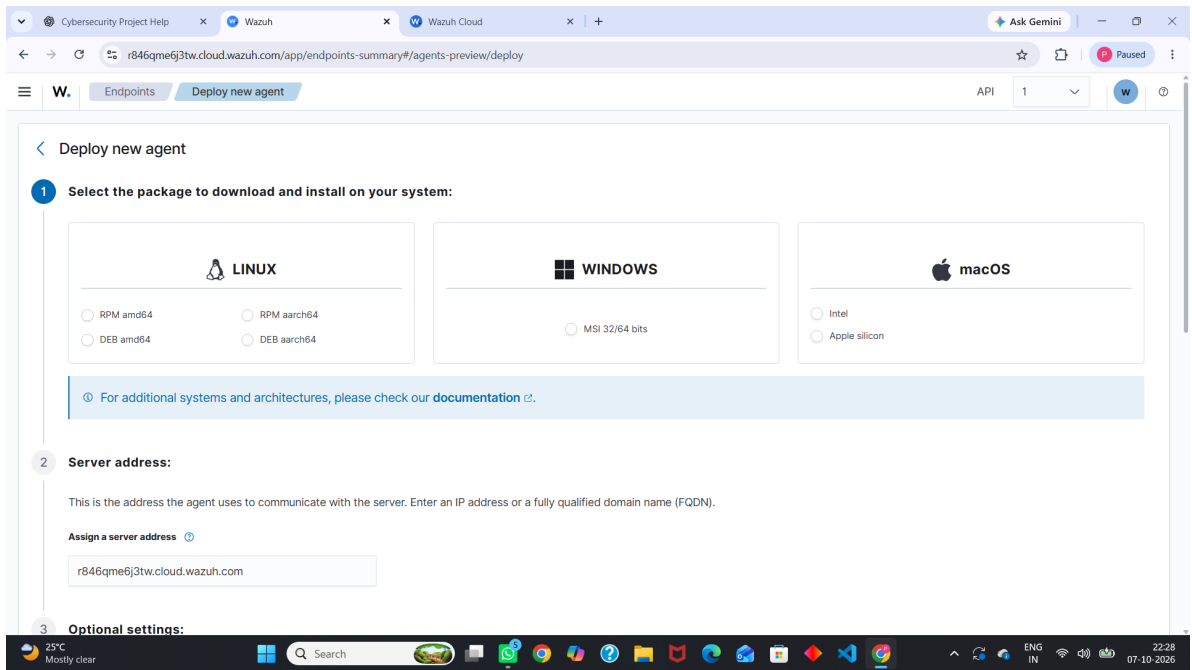
Capstone requirement	Final state
Network diagram / application map	Documented in architecture section; application access verified.
Asset inventory	Completed.
CIA analysis	Completed.
Risk assessment	Completed.
Wazuh dashboard + active agent	Completed and evidenced for WSL agent.
Nmap discovery	Completed and saved.
All Week 2 attack exercises	Not all verified; no fabricated results.
Detection notes for each activity	Target-side detection not visible in available evidence.
Incident timeline	Assessment conclusion and required next investigation steps documented.
Recommendations	Completed.
Executive summary	Completed.
Presentation	Provided as companion PowerPoint.

14. Evidence Gallery

The following pages contain representative screenshots captured during the lab build and monitoring setup.



Wazuh Cloud SecureCorp-Capstone environment.



Wazuh agent deployment wizard for Linux.

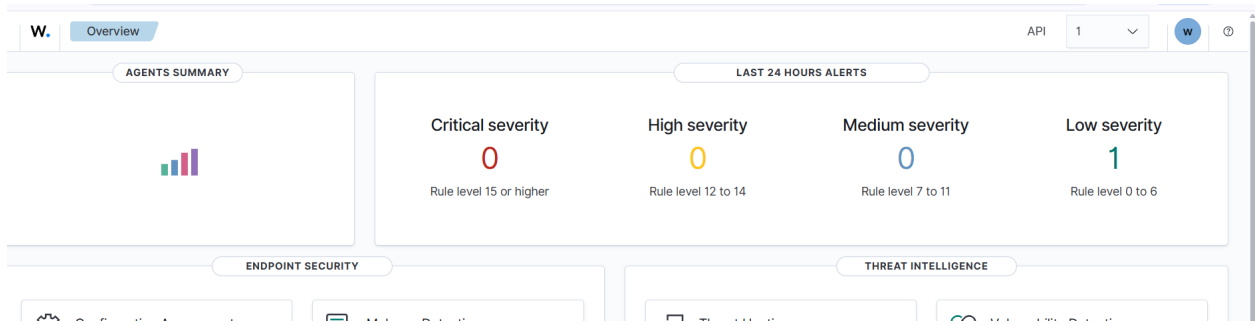
```
praneeth@praneethsurya:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.8-1_amd64.deb
&& sudo WAZUH_MANAGER='r846qme6j3tw.cloud.wazuh.com' WAZUH_REGISTRATION_PASSWORD='$'LUmDuI1tEUZzhYrDORdwk1SvGQfmcVLp' WAZUH_AGENT_GROUP='default' WAZUH_AGENT_NAME='SecureCorp-WSL' dpkg -i ./wazuh-agent_4.14.8-1_amd64.deb
--2026-10-07 17:03:06-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.8-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 108.159.80.89, 108.159.80.123, 108.159.80.65, ...
Connecting to packages.wazuh.com (packages.wazuh.com)[108.159.80.89]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13227800 (13M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.14.8-1_amd64.deb'

wazuh-agent_4.14.8-1_amd64.de 100%[=====>] 12.61M 12.6MB/s in 1.0s

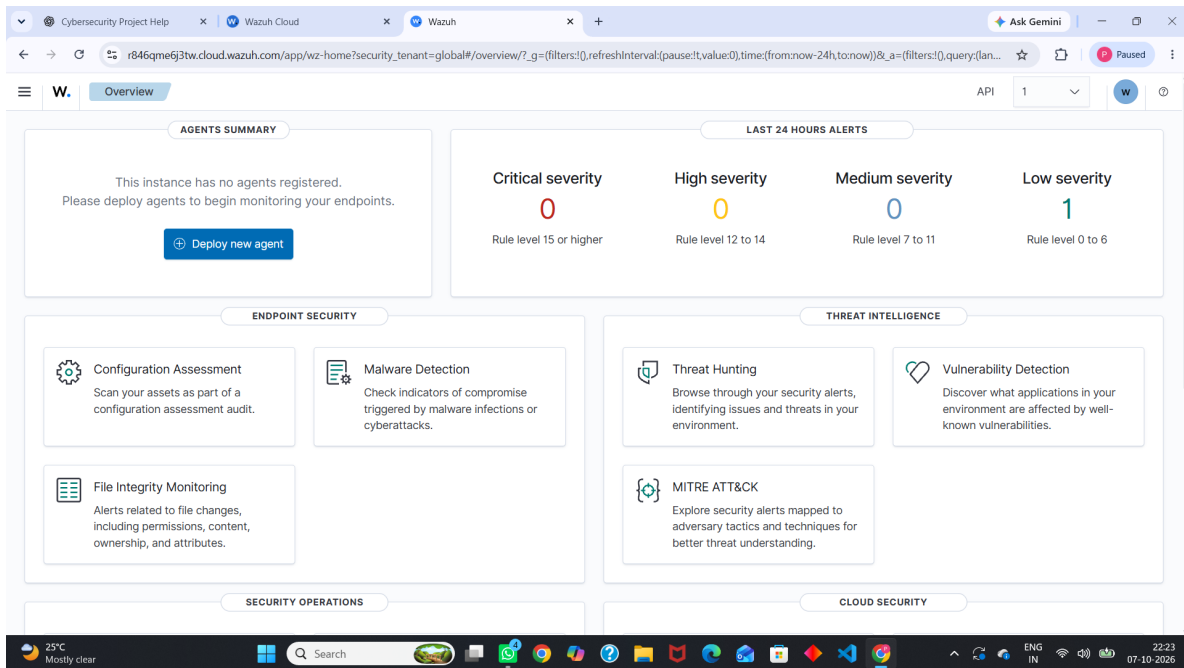
2026-10-07 17:03:07 (12.6 MB/s) - 'wazuh-agent_4.14.8-1_amd64.deb' saved [13227800/13227800]

[sudo: authenticate] Password:
Selecting previously unselected package wazuh-agent.
(Reading database ... 38983 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.14.8-1_amd64.deb ...
Unpacking wazuh-agent (4.14.8-1) ...
Setting up wazuh-agent (4.14.8-1) ...
praneeth@praneethsurya:~$
```

Wazuh agent installation in Ubuntu WSL.



Wazuh overview/dashboard evidence.



Wazuh monitoring overview captured during setup.

```
eth0    Link encap:Ethernet  HWaddr 08:00:27:9f:d8:19
        inet addr:192.168.105.48  Bcast:192.168.111.255  Mask:255.255.248.0
        inet6 addr: fe80::a00:27ff:fe9f:d819/64 Scope:Link
        UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
        RX packets:346256 errors:642 dropped:0 overruns:0 frame:0
        TX packets:2111 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:22010743 (20.9 MB)  TX bytes:203542 (198.7 KB)
        Interrupt:9 Base address:0xd020

lo      Link encap:Local Loopback
        inet addr:127.0.0.1  Mask:255.0.0.0
        inet6 addr: ::1/128 Scope:Host
        UP LOOPBACK RUNNING  MTU:16436  Metric:1
        RX packets:279 errors:0 dropped:0 overruns:0 frame:0
        TX packets:279 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:110341 (107.7 KB)  TX bytes:110341 (107.7 KB)

msfadmin@metasploitable:~$ sudo service ssh status
sudo: password for msfadmin:
sudo: service: command not found
msfadmin@metasploitable:~$ sudo /etc/init.d/ssh status
* Usage: /etc/init.d/ssh {start|stop|reload|force-reload|restart|try-restart}
msfadmin@metasploitable:~$
```

Metasploitable2 network/service troubleshooting evidence.